Exploring the Spatially Heterogeneous Effects of Urban Built Environment on Road Travel Time Variability

Shaopeng Zhong[1]; Zhong Wang[2]; Quanzhi Wang[3]; Ao Liu[4]; and Jianqiang Cui[5]

Abstract: Most studies of road travel time estimation have been based on traffic-flow theory or data-driven methods and generally neglect the influence of urban built environment on road travel time. A global regression model and a geographically weighted regression model were thus established to analyze the spatial heterogeneity of the effects of urban built environment on road travel time. The estimated results of the global regression model indicate that the occupancy rate of taxis, the distance from the nearest intersection, and the speed limit show positive correlations with a road's travel speed, whiles the number of bus stops and the distance from the nearest school show negative associations with the travel speed of the road. Furthermore, based on the results of the geographically weighted regression model, the spatially varying relationships between urban built environment and road travel time can be established, thus providing important information for decision-makers to reduce road travel time by adjusting the attributes of urban built environment. **DOI: 10.1061/JTEPBS.0000469.** *© 2020 American Society of Civil Engineers.*

Author keywords: Urban built environment; Road travel time variability; Spatial heterogeneity; Geographically weighted regression (GWR) model; Floating car data.

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Introduction

In recent years, with increasing awareness of the negative effects of travel time on transport network efficiency, road travel time estimation has attracted increasing attention in the field of intelligent transport systems (Liu et al. 2005; Li et al. 2006; Liu and Ma 2009; Jenelius and Koutsopoulos 2013; Zheng and Zuylen 2013; Pirc et al. 2016; Rahmani et al. 2017; Tang et al. 2018; Yu et al. 2018; Karimpour et al. 2019). Road travel time refers to the total time required for a vehicle to pass along a designated route (from point to point), including running time on the road, stop time, queuing delay time, and intersection delay time. Excessive road travel time not only increases monetary costs for travelers but also greatly reduces the operating efficiency of the whole road network, thereby further aggravating the jamming of the urban lifeline. Most studies of road travel time estimation have been based on data-driven methods or traffic-flow theory (Mori et al. 2014; Rahmani et al. 2015; Karimpour et al. 2019). Methods based on traffic-flow theory can be further divided into traffic delay models (van Lint 2004; Dell'Orco et al. 2016; Jiang et al. 2016), queuing theory (Liu et al. 2005), and simulation models (Liu and Ma 2009). These methods focus on transport-related factors and neglect other key influences on road travel time.

An understanding of the key factors that influence road travel time is an essential basis for research into road travel time reduction. Urban land is spatially separated into zones of diverse use, which gives rise to different types of transport demand. The separation of workplaces and residences results in commuting travel demand, and the spatial distance between residences and shopping areas causes service travel demand. In addition, features of urban built environment, as variables that characterize the spatial attributes of various urban regions, have an important impact on inhabitants' travel behavior (Fan and Khattak 2008; Wang et al. 2011; Sun et al. 2018; Liu et al. 2020). This impact mainly manifests in two aspects: (1) the layout of various types of land use in urban spaces, such as places of work, residences, and shopping areas, determines travel destinations and travel frequency; and (2) features of urban built environment, such as street design, public transport availability, and destination accessibility, may also influence the residents' choice of travel mode and travel routes (Cervero and Kockelman 1997). From this perspective, urban built environments have an impact on residents' travel behavior at the source; it influences traffic-flow distribution on the road network and ultimately affects road travel time. Therefore, establishing a relationship between urban built environment and road travel time from the perspective of urban land use and transport integration provides a feasible direction for exploration of the key factors that influence road travel time.

Most studies have focused on the influence of urban built environment on residents' travel behavior (Cervero and Kockelman 1997; Ewing and Cervero 2001; Cao et al. 2009; Ding et al. 2014). Further study is required to uncover the inherent relationship between urban built environment and road travel time and identify the key factors that affect the latter from the perspective of the former. Moreover, the complexity of urban built environment makes its impact on road travel time spatially heterogeneous (i.e., its

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impact varies in different areas). The study of the spatially heterogeneous effects of urban built environments on road travel time has important planning and practical implications; ignoring such spatial heterogeneity may result in the loss of a significant amount of detailed information about variables, leading to a biased result. Unlike global regression analysis, which can only determine the average relationships among factors, spatial heterogeneity analysis can help decision-makers identify the local relationship between urban built environment and road travel time, which is crucial to determine the causes of traffic delays in local areas.

This study aims to fill the research gap in the spatial heterogeneity analysis of the impact of urban built environment on road travel time. A global regression model and a geographically weighted regression (GWR) model are used to quantitatively address this issue based on three sources of data obtained in Shenzhen City: floating car (taxi) global positioning system (GPS) data, land use data, and road network data. To the best of the authors' knowledge, this is a pioneering study to explore the key factors that influence road travel time while taking urban built environment into consideration and the study to analyze the spatial heterogeneity of the impact of urban built environments on road travel time.

The rest of this study is organized as follows. The next section reviews the literature, after which the case study is introduced and the global regression model and GWR model are presented. Subsequently, the estimated results and the practical implications of the two models are outlined. Lastly, the conclusions of the study are discussed.

Literature Review

Urban Built Environment and Travel Behavior

The urban built environment refers to human-made environments that provide the required space for human activities, specifically including interactive spatial environments characterized by multiple factors such as land use, transport infrastructure, and urban design (Handy et al. 2002). In recent decades, scholars have paid increasingly close attention to the relationship between urban built environment (also known as urban form or neighborhood type) and travel behavior, motivated by the sprawl of car-oriented cities and increasingly serious problems (such as traffic jams, environmental pollution, and infrastructure waste) that are caused by reliance on private cars. Cervero and Kockelman (1997) and Ewing and Cervero (2001) found that travel behavior was closely related to the attributes of urban built environments, which ultimately influence road travel time by affecting citizens' travel behavior, such as travel destination, mode, frequency, and route. Studies have found that urban built environment has an important impact on car use (Maat and Timmermans 2009; Ding et al. 2014), the running mileage of cars (Cervero and Jin 2010), and the effects of the implementation of travel demand-management policies (Zhong et al. 2015; Zhong and Bushell 2017). However, no studies have focused on the influence of urban built environment on road travel time.

In terms of research methods, global regression models are the most common choice for investigating the relationship between urban built environment and travel behavior. However, because such models explore the average impact of urban built environment on residents' travel behavior across the entire spatial area under study, they can only solve spatially homogeneous problems. In reality, although samples from the same region may have certain similarities in travel behavior because of the common attributes of urban built environment, they may differ from those of other regions due to spatial heterogeneity. Neglecting such heterogeneity in research can often result in the loss of much detailed information about the relationships among variables, and the conclusions obtained will be biased.

Research on Spatial Heterogeneity

The term spatial heterogeneity is widely used to describe the inhomogeneity and complexity of spatial distributions in ecological processes and patterns. According to Anselin (1988), spatial heterogeneity refers to the fact that the characteristics of objects and phenomena in certain regions differ from those in other regions. In specific models, these differences are presented by variables, parameters, and error terms associated with different regions. To identify spatial heterogeneity, the GWR model proposed by Fotheringham et al. (2002), which uses local regression analysis and variable parameters based on local smoothing polynomials, has been widely used in many research fields, such as ecogeography, economics, medicine, and transport (Wang and Khattak 2011; Elldér 2014; Chiou et al. 2015; Feuillet et al. 2015).

It should be noted that the traditional fixed-bandwidth GWR model has two drawbacks: (1) instability occurs if the coefficients of the model suffer from collinearity of explanatory variables (Wheeler and Tiefelsdorf 2005; Páez et al. 2011); and (2) inflexibility occurs if the model assumes the same degree of spatial variation across each set of coefficients (Murakami et al. 2019). Regarding the first limitation (instability), Murakami et al. (2019) found that compared with the fixed-bandwidth GWR model, the adaptive-bandwidth GWR model effectively alleviated the instability of model results caused by changes in spatial scale. In addition, when the sample size is large, the GWR model is robust to high levels of multicollinearity (Oshan and Fotheringham 2018). However, for the second limitation (inflexibility), Murakami et al. (2019) found that for small-scale spatial variations, even the adaptive-bandwidth GWR model is vulnerable to reduced estimation accuracy.

The impacts of urban built environments on road travel time differ between regions, evidencing spatial heterogeneity. A quantitative description of this heterogeneity can contribute to a better understanding of the spatial impact of urban built environments on road travel time. In recent years, scholars have used the GWR model to solve complicated spatially heterogeneous problems in transport research, such as estimation of average daily traffic on a nonhighway road (Zhao and Park 2004), discovery of the relationship between transport accessibility and land value (Du and Mulley 2006), exploration of the effects of travel information on travel decisions (Wang and Khattak 2011), study of the spatial variations in the relationships between the number of crashes and explanatory variables (Pirdavani et al. 2014; Ariannezhad et al. 2020), identification of the key factors that influence the utilization rates of public transport (Chiou et al. 2015; Ma et al. 2018), evaluation of the sensitivity of on-street parking occupancy to price changes (Pu et al. 2017), and examination of the effect of transportation accessibility on regional economic resilience (Chacon-Hurtado et al. 2020). Although the GWR model has been validated as an effective approach to study spatial heterogeneity in the transport field, few studies have focused on the relationship between urban built environment and road travel time.

Research on Road Travel Time

Most studies of road travel time estimation have been based on traffic-flow theory or data-driven methods (Mori et al. 2014; Rahmani et al. 2015; Karimpour et al. 2019). Traffic-flow theory

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is used to estimate road travel time through traffic delay models, queuing theory, and simulation models. These methods consider various traffic parameters in the road network and factors that cause traffic delay (Vanajakshi 2005; Hellinga et al. 2008; Liu and Ma 2009; Dell'Orco et al. 2016). For example, van Lint (2004) used such a method to predict the traffic conditions in future time intervals and estimated the travel time from the predicted traffic conditions and variables. Liu et al. (2005) used a traffic delay model and queuing theory, which are traditional traffic-flow theories, to predict road travel time. In contrast, unlike traffic-flow theory, data-driven methods process large amounts of data using informatics methods such as linear regression, Bayesian networks, artificial neural networks, Kalman filtering, fuzzy algorithms, and machine learning or deep learning and thus avoid the difficulties of relying on strict mathematical derivations (Lee et al. 2006; Zheng and Zuylen 2013; Rahmani et al. 2015; Ma et al. 2017; Tang et al. 2018; Xu et al. 2020).

Elsewhere, other researchers have combined traffic-flow theory with data-driven methods to estimate road travel time (Hofleitner et al. 2012). However, whether these studies are based on traffic-flow theory, data-driven methods or combined methods, they have mostly paid attention to road parameters and historical travel time data, while usually neglecting the factors of urban built environment that influence road travel time. It is important to clarify the relationship between the attributes of urban built environment and road travel time, because the latter can be minimized by adjusting the former.

This study aims to contribute to the existing literature in the following aspects:

1. Identify the key factors of urban built environment that influence road travel time using a global regression model with interaction terms;
2. Incorporate the impact of the occupancy rate of taxis when using taxi GPS data to estimate road travel time;
3. Analyze the spatial heterogeneity of the relationships between urban built environment and road travel time using a GWR model; and
4. Provide important information for transport planning and practical decision-making to reduce road travel time by adjusting the attributes of urban built environment.

Case Study

Selected Study Region

The target route is situated in Nanshan District, Shenzhen, starting from the intersection of the Industrial 8th Road and Houhai Road and ending at the intersection of the Qiaocheng East Road and Baishi Road. This route passes through 17 intersections with a total length of approximately 10 km. The study route and the intersection names are presented in Fig. 1, which is modified from Google Maps, where C1–C17 refer to intersections 1–17.

Data Collection

Taxi GPS data, land use data, and road network data of Shenzhen were collected. Depending on road operational conditions, the taxi GPS data were uploaded at a frequency of approximately three or four times per minute. Along the study route (Fig. 1), 8,546 records of taxi GPS data, including license number, time, latitude and longitude, driving direction, speed, and occupancy state of the taxi, were extracted from 1,302 taxis in a 2-h period from 7:30 to 9:30 a.m. on 3 days (i.e., from Tuesday, June 10 to Thursday, June 12, 2014). The land use data geographically represent different types of land use in Shenzhen, consisting of the spatial positions and attributes of supermarkets, schools, hospitals, hotels, and other buildings. The road network data comprise geographic information of the Shenzhen road network and data on Shenzhen from Baidu Maps.

After collecting the required data, ArcGIS version 10.2 was used to divide the research route into 397 road segments of 25 m each. Each segment was treated as an independent sample. The average speed of each road segment and the occupancy rate of taxis were then obtained by screening and matching the taxi GPS data. Next, ArcGIS was used to analyze the land use data

Fig. 1. Study route. (© Google.)

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within a radius of 500 m around the study route to obtain the numbers of buildings of each type and public facilities around each road segment. Preliminarily, three different ranges were tested for this buffer around the road segments: 200, 500, and 1,000 m. Considering the requirements for statistical significance and the interpretability of the model, 500 m was chosen. Finally, to elucidate the interactive effects of the intersection types and urban built environment on road travel time, dummy variables were added into the model. These dummy variables were used to classify the intersection types in three respects: the number of entrance lanes, presence or absence of left-turn lanes, and whether the left-turn lanes are independent.

Average Speed and Occupancy Rate of Taxis

For a fixed road distance, the road travel time varies inversely with the average travel speed. Therefore, the road travel time was represented by the average travel speed of a road segment and calculated by screening, correcting, and matching the taxi GPS data (Wolf et al. 2004). The use of taxi GPS data to estimate the travel time of the road segments is subject to some sources of error, which may lead to over or underestimated values. Specifically, when the occupancy rate of taxis on a road segment is high, the travel time of the segment is usually overestimated because occupied taxis usually travel faster than normal private vehicles. Conversely, when the occupancy rate of taxis on a road segment is relatively low, the travel time of the segment is, in general, underestimated, because unoccupied taxis usually travel more slowly than normal private vehicles. To calculate and control the impact of the occupancy rate of taxis on road travel time, the occupancy rate of taxis was added as an independent variable. Note that if data were available, the use of a broader database (i.e., one that contains all floating car data) would also alleviate this biased estimation problem.

Measurements of Urban Built Environment

For this study, 14 measurements of an urban built environment were selected: the number of office buildings, banks, hotels, pharmacies, car parks, supermarkets, restaurants, bus stops, and schools; the distances (Euclidean distance between a road segment and the centroid of the objective) from the nearest school, from the nearest intersection (intersection on the study route) and from the nearest bus stop; the speed limit within a range of 500 m around the study route; and the geometric design attributes of the roads (number of lanes in one direction). Each road segment has the same length (25 m); thus, the number of facilities reflects the density of facilities.

Classification of Intersection Types

When studying the impact of other variables on road travel time, the type of intersection must be considered because different intersection types often have different impacts. The 17 intersections on the study route were divided into four types according to the number of entrance lanes, presence or absence of left-turn lanes, and whether the left-turn lanes are independent. The resulting classification and a schematic diagram of each class of intersection are shown in Table 1 and Fig. 2, respectively. An independent left-turn means that the intersection has an independent zone in which the left-turning traffic can wait. For example, in Fig. 2, intersection Types 1 and 4 have independent left-turn lanes.

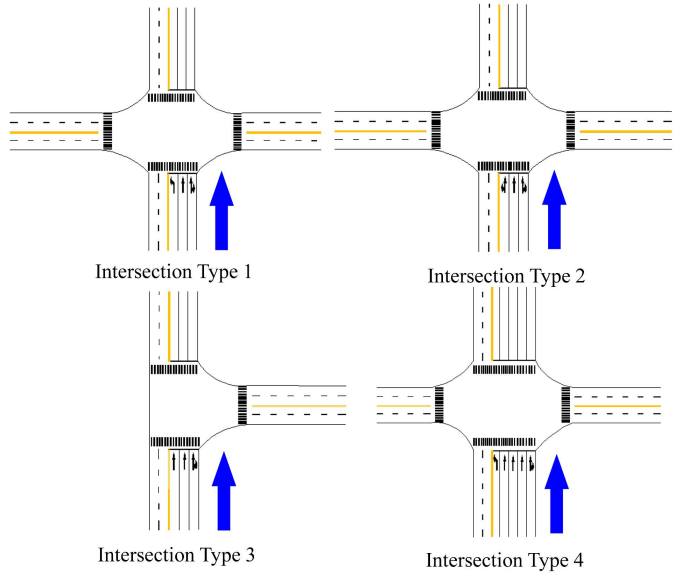

Fig. 2. Schematic diagram of four intersection types.

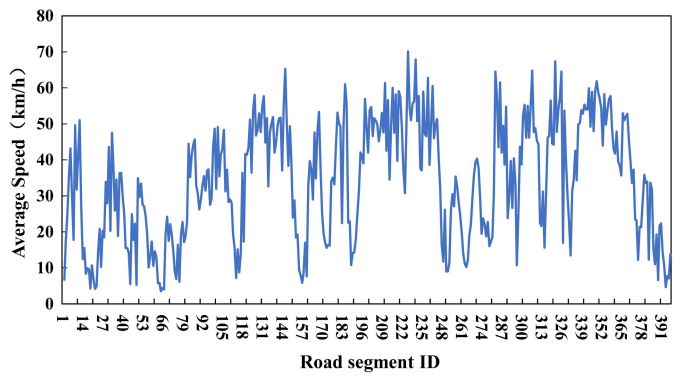

Fig. 3. Distribution of the average travel speed on each road segment.

Descriptive Statistics of Variables

In this study, the dependent variable is the average travel speed of a road segment. Fig. 3 presents the distribution of average travel speed on each of the 397 road segments. The average travel speed is greatly affected by delays at intersections, so the average travel speeds at all of the intersections (with a maximum value of 22.72 km/h) are clearly lower than that of the research route as a whole (34.32 km/h).

Table 1. Classification of intersection types

Intersection types	Features	Classification result
Type 1	The number of entrance lanes does not exceed four, and there are independent left-turn lanes	C2, C5, C7, C8, C9, C11, C12
Type 2	The number of entrance lanes does not exceed four, and there are left-turn lanes but they are not independent	C3, C16
Type 3	The number of entrance lanes does not exceed four, and there is no left-turn lane	C13, C14, C15
Type 4	The number of entrance lanes exceeds four, and there are independent left-turn lanes	C1, C4, C6, C10, C17

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Table 2. Descriptive statistics of the independent variables

Variable	Unit	Minimum value	Maximum value	Average value	Standard deviation
Number of office buildings	count	0	8	1.68	2.41
Number of banks	count	0	33	7.21	8.25
Number of hotels	count	0	8	1.20	1.90
Number of pharmacies	count	0	21	4.25	5.52
Number of car parks	count	0	20	4.31	4.94
Number of supermarkets	count	0	126	22.39	31.41
Number of restaurants	count	0	75	16.26	19.45
Number of bus stops	count	0	38	13.67	8.89
Number of schools	count	0	19	4.35	5.27
Distance from the nearest school	m	8	995	350.43	282.27
Distance from the nearest intersection	m	0	725	202.08	164.08
Distance from the nearest bus stop	m	0	575	195.03	140.02
Speed limit	km/h	50	60	54.56	4.99
Number of lanes in one direction	count	3	4	3.14	0.34
Occupancy rate of taxis	—	0	1	0.57	0.19

In the global regression model with interaction terms, the attributes of urban built environment and the occupancy rate of taxis were set as the independent variables. The descriptive statistics of the independent variables are provided in Table 2. In addition, the interaction between the intersection types and the independent variables was considered. Dummy variables were introduced to specifically quantify the impact of intersection types on road travel time, and intersection Type 4 was chosen as a reference item. In summary, the global regression model included 59 initial independent variables, including independent variables, dummy variables, and interaction terms between them.

Model Formulation

In this section, a global regression model is first established. The model is designed not only to extract the main independent variables that influence road travel time but also to reveal whether the effects of urban built environments on road travel time are spatially heterogeneous. The spatial heterogeneity of the effects of urban built environment on road travel time is then analyzed using the established GWR model.

Global Regression Model

The impact of the attributes of urban built environment on the average travel speed of a road segment is affected by the type of the nearest intersection. Therefore, in the global regression model, the type of the nearest intersection to the road segment is set as the regulating variable to analyze the interaction between them. Because the road travel time is equal to the length of the road divided by the average travel speed, the average travel speed is adopted as the dependent variable instead of the road travel time. Specifically, in the global regression model, the average travel speed of a road segment is set as a dependent variable, the attributes of urban built environments and the occupancy rate of taxis are set as independent variables, and the dummy variables of the nearest intersection type to the road segment are set as regulating variables. For the interaction under the framework of a regulating variable, the effect of the independent variables on the dependent variable depends on the value of the regulating variable. The structure of the regression model is formulated as

$$s = \beta_o + \sum_{k=1}^{m} \beta_k x_k + \sum_{p=1}^{n} \eta_p D_p + \sum_{k=1}^{m} \sum_{p=1}^{n} \lambda_{kp} x_k D_p + \varepsilon \qquad (1)$$

where s = average travel speed of a road; β_o = constant; x_k represents an attribute of the built environment, which is an independent variable (Table 2); β_k represents the regression coefficient corresponding to x_k; m = number of independent variables; D_p = nearest intersection type to the road segment, which is a regulating variable; η_p represents the regression coefficient corresponding to D_p; n = number of regulating variables; λ_{kp} = coefficient of the impact of the interaction between independent variables and regulating variables; and ε = random error.

The global regression model can be estimated based on the ordinary least-squares method, which minimizes the square of the distance between the observed values and the estimated values, namely, the residual sum of squares. The results of the global regression model reveal whether an interaction lies between the independent variables and the regulating variables.

GWR Model

In the global regression model, the spatial heterogeneity of variables cannot be captured, because all of the differentiated spatial units are represented by mean values. This can lead to a great loss of detailed information, causing the obtained result to be biased. Unlike the global regression model, in the GWR model, the estimated parameters of each unit vary with location. This allows the structural relationships among variables in various locations and the spatial heterogeneity to be easily established. The spatial locations of variables are embedded in the regression parameters, which can be estimated using the locally weighted least-squares method. The structure of the GWR model is formulated as follows:

$$s_i = \beta_{i0} + \sum_{k=1}^{m'} \beta_{ik} x_{ik} + \varepsilon_i \qquad (2)$$

where s_i = average speed of road segment i; β_{i0} = constant of road segment i; x_{ik} represents the kth independent variable associated with road segment i; β_{ik} = regression coefficient corresponding to x_{ik}; m' = number of independent variables that are statistically significant in the global regression model; and ε_i = random error of road segment i.

When estimating the parameters of road segment i using only its own set of data, the result is unbiased, but the standard deviation of the result is large, and the accuracy is poor. To remedy this problem, more samples around road segment i must be added. A consequent problem is that if adjacent samples of road segment i are added, some extent of deviation is inevitable, and this deviation

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increases with the number of samples. To balance the deviation of the estimated parameters in the GWR model against the standard deviation, it is necessary to select appropriate adjacent samples.

To determine which sample road segments adjacent to road segment i should be used for estimation of model parameters, a spatial weight matrix $\mathbf{W}(i)$ is introduced to give a spatial weight value to each adjacent segment of road segment i. This establishes a function of the distance between the regression segments and the other sample segments through a spatial weight function. In practice, the spatial distribution of sample points tends to be uneven. Some areas are dense in sample points, while others are sparse. The study route includes both straight and curved road segments. Because the density of curved road segments is higher than that of straight road segments, if the bandwidth is fixed in the weight function, more sample road segments will be selected in the areas with high sample density, while fewer will be selected in the areas with low sample density. In response to this problem, the adaptive-bandwidth method is adopted to assign different optimal bandwidth values to regions with different sample densities. In the case of adaptive bandwidth, the number of road segments included in the bandwidth is kept constant, making the bi-square weight function more suitable for the Gaussian weight function (Oshan et al. 2019). The expression of the bi-square weight function is as follows:

$$\mathbf{W}_j(i) = \begin{cases} [1 - d_{ij}^2/b_i^2]^2, & d_{ij} \le b_i \\ 0, & d_{ij} > b_i \end{cases} \tag{3}$$

where $\mathbf{W}_j(i)$ = spatial weight of road segment j adjacent to road segment i; d_{ij} = distance between road segments i and j; and b_i = bandwidth of road segment i. Then, the regression parameter matrix $\hat{\boldsymbol{\beta}}(i)$ at road segment i is estimated as follows:

$$\hat{\boldsymbol{\beta}}(i) = [\mathbf{X}^T\mathbf{W}(i)\mathbf{X}]^{-1}\mathbf{X}^T\mathbf{W}(i)\mathbf{S} \tag{4}$$

where $\mathbf{X}$ = matrix of independent variables with $m' + 1$ columns and a column of 1s for the intercept; $\mathbf{S}$ = vector of dependent variables; and $\mathbf{W}(i)$ = weight matrix corresponding to road segment i, which is a diagonal matrix. Lloyd (2010) found that the results of GWR models were more sensitive to the bandwidth of a specific weight function than to the type of weight function. Here, the bandwidth was selected by the minimization of Akaike information criterion (AIC) (Fotheringham et al. 2002).

Results and Analysis

The average travel speed of the road segments was estimated using the global regression model and the GWR model. It is worth noting that the local estimated coefficients may not be consistent with the global estimated results, because some estimated global parameters may take the opposite sign at some locations. The results are presented in the following two sections, together with figures depicting the spatial variations.

Results and Analysis of the Global Regression Model

The global regression model was estimated using SPSS Statistics 17.0, and the results are shown in Table 3. Considering the large number of independent variables, only the statistically significant independent variables are listed in Table 3. The F-value of the model is 13.678, and the corresponding number of degrees of freedom is 61. Therefore, the null hypothesis is rejected, that is, at least one coefficient of independent variables is significantly different from 0. R_a^2 is 0.47, indicating that the independent variables in the model can explain 47% of the variation in the average speed of the road segments.

It can be seen from Table 3 that the distance from the nearest intersection, the speed limit, and the occupancy rate of taxis show positive correlations with the average speed of the road segments, while the corresponding correlations with the number of bus stops and the distance from the nearest school are negative. These results are intuitively reasonable. For example, the higher the speed limit and the further from the nearest intersection, the higher the average speed of the road segment. In addition, Table 3 also suggests that intersection Type 1 shows a positive correlation with the average speed of the road segments, which indicates that left-turn lanes are conducive to improving the speed of vehicles as they pass through an intersection.

Table 3. Estimated results of the global regression model and the GWR model

Models	Global regression model			GWR model			
Variable	Coefficient	p-value	Minimum	1st quantile	Median	3rd quantile	Maximum
Constant	−82.94	0.001	−122.11	−18.28	18.54	31.59	76.73
Number of bus stops (a)	−1.80	0.000	−0.52	−0.08	0.32	0.55	1.05
Distance from the nearest school (b)	−0.03	0.032	−0.03	−0.01	0.01	0.03	0.04
Distance from the nearest intersection (c)	0.04	0.001	0.04	0.05	0.07	0.07	0.08
Speed limit (d)	1.93	0.000	−0.91	−0.23	−0.08	0.79	2.33
Occupancy rate of taxis (e)	28.22	0.000	−4.88	5.91	11.92	16.31	22.00
Dummy							
Intersection Type 1 (D1)	323.85	0.000	—	—	—	—	—
Interaction term							
a × D1	2.48	0.000	—	—	—	—	—
a × D2	2.87	0.007	—	—	—	—	—
a × D3	2.52	0.004	—	—	—	—	—
b × D1	0.03	0.039	—	—	—	—	—
c × D1	0.05	0.000	—	—	—	—	—
c × D3	0.03	0.051	—	—	—	—	—
d × D1	−4.88	0.000	—	—	—	—	—
d × D2	−3.75	0.019	—	—	—	—	—
e × D1	−19.12	0.010	—	—	—	—	—
R^2	0.48				0.59		
R_a^2	0.47				0.56		
AIC	3,113.91				3,050.34		

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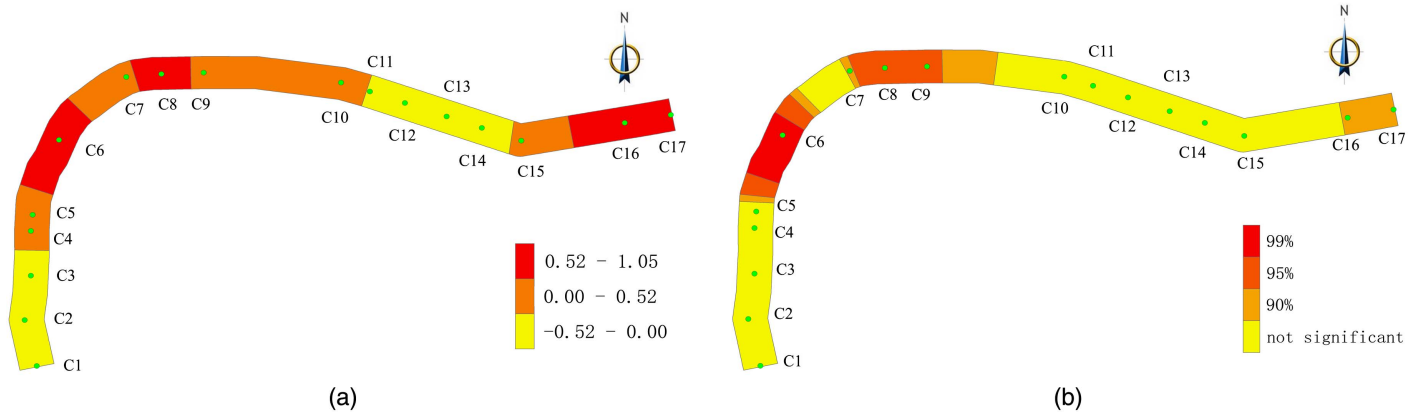

Fig. 4. (a) Spatial distribution of the regression coefficients of the number of bus stops; and (b) spatial distribution of the p-values of the number of bus stops.

Inspection of the interaction terms verifies that interaction does exist between the intersection type and urban built environment and between the intersection type and the occupancy rate of taxis. Another noteworthy finding is that factors that have a significant influence on the average speed of the road segments vary with location. This indicates that the influence of urban built environment on the average speed of the road segments is not uniform across the entire research route, because the type of the nearest intersection varies for different road segments, which leads to spatial heterogeneity. However, the global regression model only summarizes the overall pattern, and the result is the average value obtained on the entire route. Therefore, the GWR model is necessary to reveal the underlying spatial heterogeneity.

Estimated Results and Analysis of the GWR Model

Spatial heterogeneity refers to the differences between objects and phenomena in a certain geographic position from those in other regions (Anselin 1988; Fotheringham et al. 2002). Analysis of the interaction terms in the global regression model verifies that the effects of urban built environment on road travel time are spatially heterogeneous. Therefore, the GWR model is used to study this spatial heterogeneity and the reasons for its existence.

Model Estimation

The independent variables used in the GWR model were determined according to the result of the global regression model. In addition, the adaptive-bandwidth method was used, wherein the optimum bandwidth was determined by minimizing the AIC value. The impact of each road segment's spatial location on its regression coefficient was described by a bi-square weight function. The MGWR version 2.2.1 software package was used to estimate the GWR model (Oshan et al. 2019). The results are the corresponding regression coefficients of the 397 road segments. Due to space limitations, only the minimum value, first quantile value, median, third quantile value, and maximum value of the regression coefficients for each independent variable are shown in Table 3.

Table 3 shows that the same independent variable has different effects on the average speed of various road segments. Specifically, some independent variables show positive correlations with the average speed on some road segments, but negative correlations with the average speed on other road segments. Therefore, it is necessary to analyze the spatial influence of each independent variable on the average speed of the road segments.

Variable Analysis

The following analysis focuses on five factors: the number of bus stops, the distance from the nearest school, the distance from the nearest intersection, the occupancy rate of taxis, and the speed limit.

Number of Bus Stops. Fig. 4 shows the spatial distribution of the regression coefficients and the p-values of the number of bus stops on the study route. A noteworthy finding is that the number of bus stops has a positive correlation with the average speed of the road segments between intersections 5 and 6, between intersections 7 and 9, and between intersections 16 and 17. Although the total length of the three segments is less than 40% of the entire study route, 60% of the bus stops are located within 500 m of these segments. In all three segments, the average speed is sensitive to the number of bus stops; specifically, the greater the number of bus stops, the shorter the road travel time. This relationship has two causes. First, the taxi GPS data were collected during the hours of exclusive bus lane service (7:30–9:30 a.m.). Therefore, despite a large number of bus stops on these road segments, the bus stops do not have a negative impact on the speed of private vehicles due to the exclusive bus lanes and the harbor-shaped bus stops, which segregate buses from other vehicles. Second, the more bus stops there are, the higher the probability of commuters traveling by bus (Sun and Guan 2016) and the lower the probability of commuters traveling by other modes (including taxi). Correspondingly, the possibility of taxis decelerating to pick up passengers is lower. Therefore, the average speed of the road segments is higher when represented by the travel speed of taxis.

Distance from the Nearest School. Fig. 5 presents the spatial distribution of the regression coefficients and the p-values of the distance from the nearest school on the study route. It can be seen that the distance from the nearest school has a positive correlation with the average speed of the road segments between intersections 3 and 7. That is, on these segments, the shorter the distance from the nearest school, the longer the road travel time. Fig. 6(a) shows that the building density on these road segments is high. Most vehicles that travel to nearby schools go through these road segments, and the entrances of some schools also face these segments. Thus, these segments are particularly sensitive to the presence of schools, such that the shorter the distance to the nearest school, the greater the impact. However, for the road segments between intersections 11 and 14, the distance from the nearest school has a negative association with the average speed of the road segments. That is, on these segments, the longer the distance from the nearest school, the

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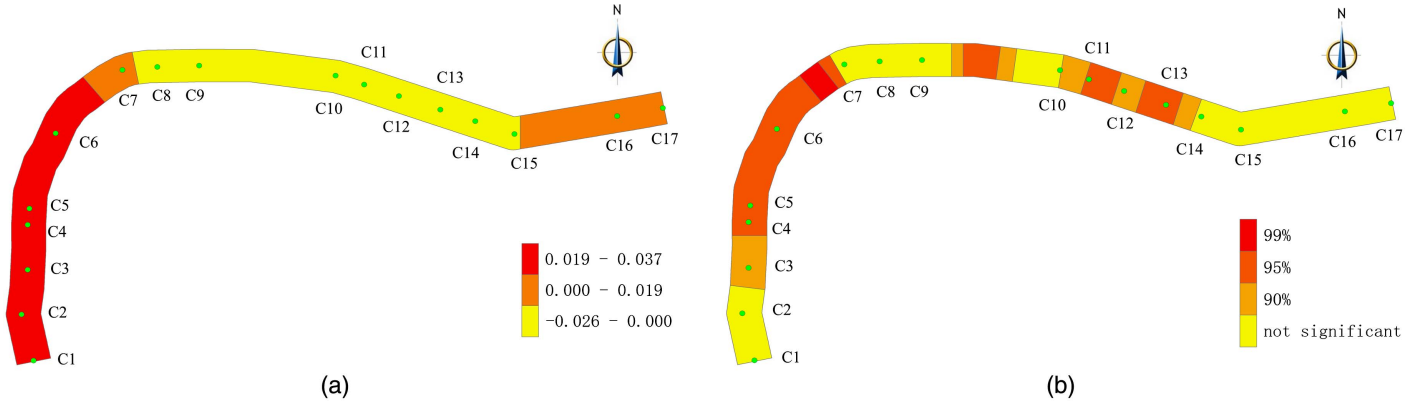

Fig. 5. (a) Spatial distribution of the regression coefficients of the distance from the nearest school; and (b) spatial distribution of the p-values of the distance from the nearest school.

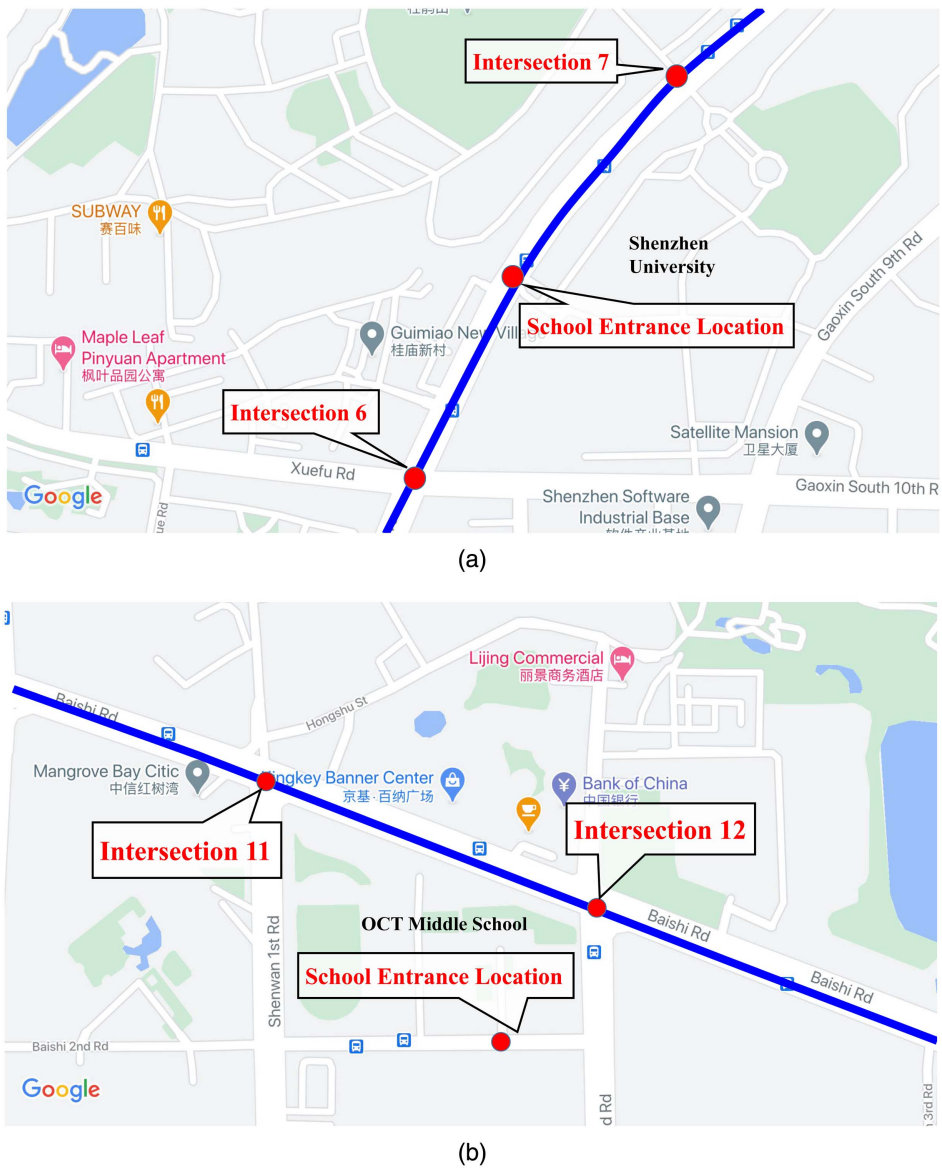

Fig. 6. (a) Road segments between intersections 6 and 7 on the study route; and (b) road segments between intersections 11 and 12 on the study route. (© Google.)

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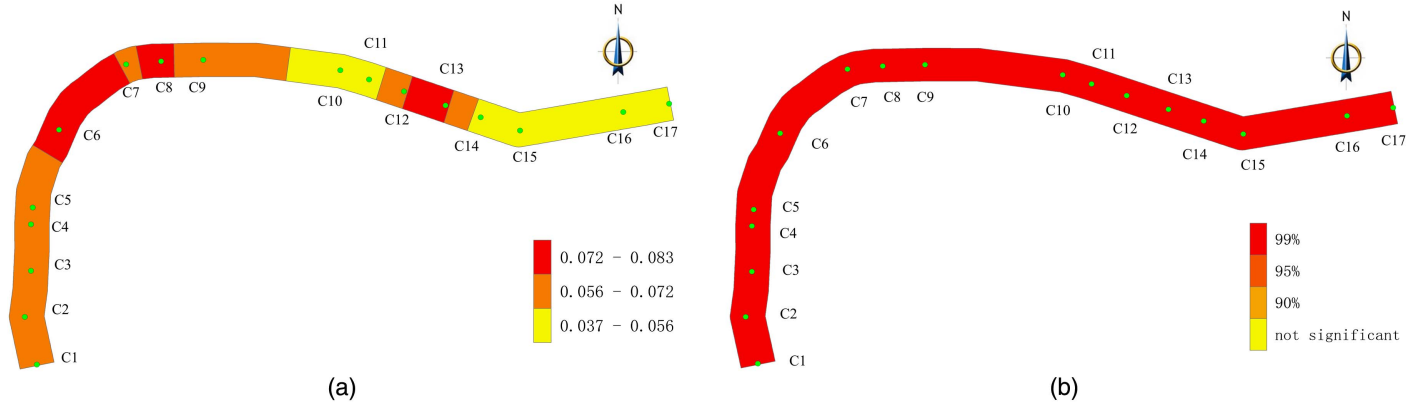

Fig. 7. (a) Spatial distribution of the regression coefficients of the distance from the nearest intersection; and (b) spatial distribution of the p-values of the distance from the nearest intersection.

lower the average speed. This is because these road segments are close to the rear side of the school, while the school entrance is on a road parallel to these segments [Fig. 6(b)]. This means that the road segments with a shorter direct Euclidean distance from the school are in fact further from the school entrance. In contrast, the road segments with a longer direct distance from the school are closer to the school entrance in terms of the actual road network. Consequently, these road segments are more likely to be affected by the pedestrian flow and the traffic flow in front of the school.

Overall, the impact of the distance from the nearest school on the average speed of the road segments is statistically significant between intersections 3 and 7 and between intersections 11 and 14. The school density is higher on these road segments than on others, and different locations of the main entrances of the nearby schools have different effects on the average speed of the segments. These results confirm that schools have an important impact on the travel time on surrounding roads, and the shorter the distance from the school entrance to the road, the greater the impact. Therefore, to reduce road travel time, schools should be built as far as possible from main urban roads. If schools are already located near the main roads, their impact can be reduced by relocating the school entrances.

Distance from the Nearest Intersection. Fig. 7 illustrates the spatial distribution of the regression coefficients and the p-values of the distance from the nearest intersection on the study route. The figure shows that the distance from the nearest intersection has a positive correlation with the average speed on the whole study route. That is, the shorter the distance from the nearest intersection, the lower the average speed of the road segments, and the longer the road travel time. By comparing road segments with respect to the nearest intersection type, it is found that if the nearest intersection is Type 1 or Type 4, the regression coefficient is relatively large, whilst if the nearest intersection is Type 2 or Type 3, the regression coefficient is relatively small. This indicates that the presence of left-turn lanes has a great impact on the average speed of a road segment. All else being equal, the average speed of a segment whose nearest intersection has an independent left-turn lane is higher.

Occupancy Rate of Taxis. Fig. 8 shows the spatial distribution of the regression coefficients and the p-values of the occupancy rate of taxis on the study route. The occupancy rate has a positive correlation with the average speed of the road segments between intersections 6 and 8 and between intersections 11 and 17. That is, the average speed of these road segments increases as the taxis' occupancy rate increases. This indicates two things. First, the frequency of passengers boarding and alighting from taxis is high on these road segments; thus, the average speed of these segments is highly dependent on the occupancy rate of taxis. Therefore, to reduce the interference of taxis with private vehicles, temporary taxi stands could be designed and constructed on these road segments to segregate taxi drivers and passengers from other vehicles. Second, the occupancy rate of taxis has a great impact on road travel time,

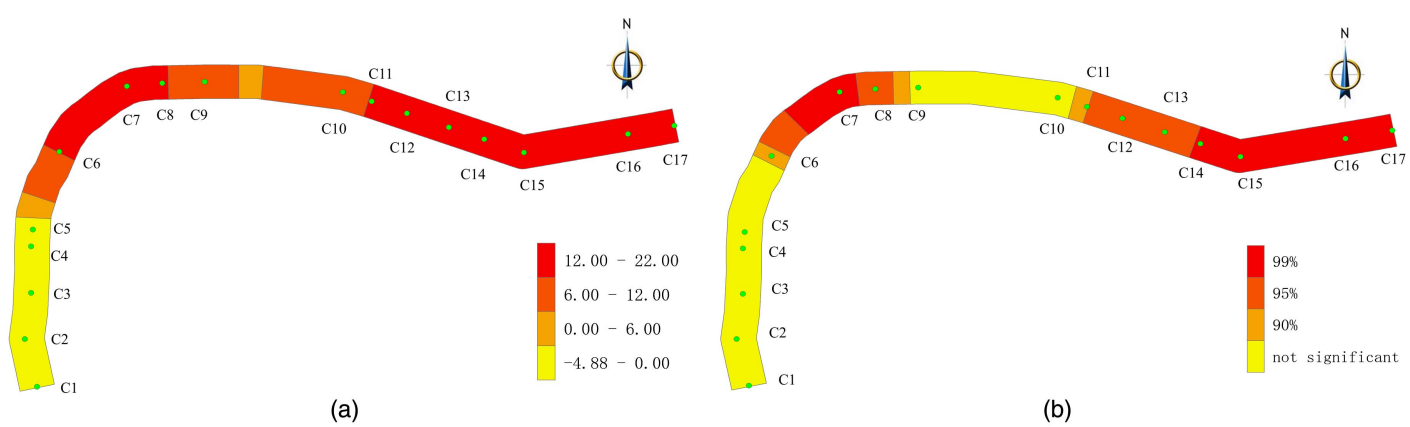

Fig. 8. (a) Spatial distribution of the regression coefficients of the occupancy rate of taxis; and (b) spatial distribution of the p-values of the occupancy rate of taxis.

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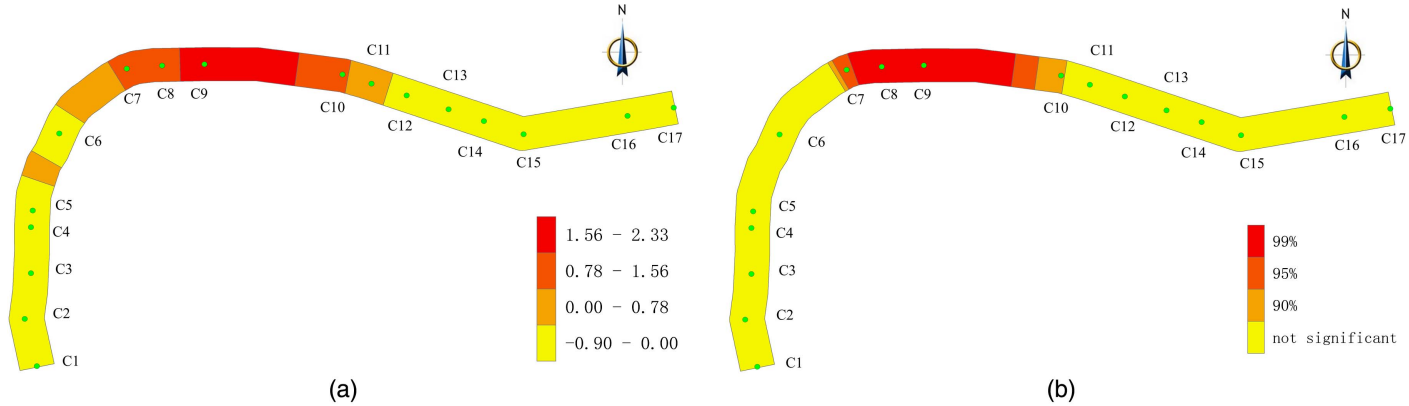

Fig. 9. (a) Spatial distribution of the regression coefficients of the rate of the speed limit; and (b) spatial distribution of the p-values of the rate of the speed limit.

because taxis may deliberately slow down and stop to pick up passengers at any time when vacant. Therefore, when estimating road travel time based on taxi GPS data, neglecting the status (i.e., occupied or empty) of taxis will lead to biased estimations. **Speed Limit**. The target route of this study is approximately 10-km long, with the speed limit ranging from 50 to 60 km/h on different sections, which may have a direct impact on the travel time of each section. Fig. 9 depicts the spatial distribution of the regression coefficients and the p-values of the speed limit on the study route. The speed limit only impacts on the average speed of the road sections between intersections 7 and 10, where the speed limit has a positive correlation with the average speed of these road sections. That is, the higher the speed limit, the higher the average speed of the road sections, and the shorter the road travel time. Generally, speed limits are set in accordance with the road conditions. Road sections capable of supporting a higher travel speed tend to have a higher speed limit. Here, the fact that the speed limit is only significant between intersections 7 and 10 indicates that that part of the road is in good condition, thus drivers are more likely to exceed the speed limit there (road sections 158–250; see Fig. 3). This finding suggests that the transport department should actively monitor speeding behavior to ensure the safety of the road network. Another action would be to set more reasonable speed limits by considering the actual conditions of each road section.

Method Application and Policy Implications

The method proposed in this study can be applied in two practical aspects. First, the method can assist transport planners and managers to better understand the relationship between urban built environment and road travel time. The traditional data-driven methods (referring to those that use quantitative data, as opposed to those driven by models or theories) cannot analyze the internal mechanism of the impact of various factors on road travel time. Overcoming this limitation of the traditional data-driven methods, this study established a general relationship between urban built environment and road travel time, which can capture the internal factors that influence road travel time. Second, the proposed method and the obtained results can help decision-makers develop targeted urban planning and management strategies. By adjusting the attributes of urban built environment (such as the type of bus stops, the location of school entrances, and the type of intersections), road travel time can be reduced.

By analyzing the spatial impact of urban built environments on the average speed of the road segments, this study provides the following important policy implications for transport planning and management. First, the coordination of exclusive bus lanes and harbor-shaped bus stops can reduce the negative effects of bus stops on road travel time, which corroborates the findings of Jia et al. (2009) and Arasan and Vedagiri (2010), Second, the impact of a school on road travel time depends mainly on the location of the school entrance, and the negative effects can be reduced by relocating the entrance, which is consistent with previous findings (e.g., Mackie 2010), Third, along the whole study route, the distance from the nearest intersection has a positive correlation with the average speed of the road segments. When the nearest intersection to the road segment has an independent left-turn lane, the corresponding regression coefficient is larger and the average speed of the segment is higher. Fourth, the occupancy rate of taxis has a great effect on road travel time. Therefore, when estimating the average travel time of the road segments based on taxi GPS data, the taxis' occupancy status must be taken into account. Finally, the speed limit has a significant positive correlation with the average speed of the road sections on some parts of the route, so the urban transport department is advised to establish a system to monitor speeding in these areas to ensure road safety.

Conclusions

Thus far, most studies on road travel time estimation have been based on traffic-flow theory or data-driven methods. These approaches take close account of traffic status and historical travel time data but largely fail to capture the key factors of urban built environment that influence road travel time. In view of this, this study presented pioneering research that identified and examined the key factors that influence road travel time by using a global regression model and a GWR model from the perspective of urban built environment.

The results of the global regression model show that the distance from the nearest intersection, the occupancy rate of taxis, and the speed limit have positive correlations with the average speeds of the road segments, while the corresponding correlations with the number of bus stops and the distance from the nearest school are negative. In addition, there is an interaction between the intersection type and the features of urban built environments and between the intersection type and the occupancy rate of taxis. Moreover, factors that significantly affect the average speed of the road (or, equivalently, the road travel time) depend on the type of intersection nearest to the road segment because of spatial heterogeneity. The GWR model results showed that the number of bus stops, the distance

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from the nearest school, the distance from the nearest intersection, the occupancy rate of taxis, and the speed limit can have either positive or negative associations with the average speeds of the road segments. These factors only have significant effects on parts of the study route.

Last, but not least, this study addressed two drawbacks of the traditional fixed-bandwidth GWR model: instability and inflexibility. By using the adaptive-bandwidth GWR model and a large sample of taxi GPS data, the impact of collinearity on the model results was effectively alleviated, ensuring the stability of the estimated results. For the second limitation (inflexibility), Murakami et al. (2019) found that small-scale spatial variations affect the estimation accuracy of the adaptive-bandwidth GWR model. Therefore, given the small spatial scale, caution should be exercised when using the results of this study. One possible future research direction is to use the flexible adaptive-bandwidth GWR model to solve this problem. Compared with the nonflexible adaptive-bandwidth GWR model, the flexible variant can not only ensure the stability of the estimated results (e.g., in the presence of collinearity), but also control the varying scales of different sets of coefficients. It is worth noting, however, that compared with the traditional GWR model, the computational efficiency of the flexible GWR model will be reduced, especially for large data sets.

Data Availability Statement

All data, models, or code generated or used during the study are confidential in nature. All these items are part of a project with the Transport Bureau of Shenzhen Municipality, so they are not allowed to be shared.

Acknowledgments

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Notation

The following symbols are used in this paper:

b_i = bandwidth of road segment i;
D_p = nearest intersection type to the road segment, which is a regulating variable;
d_{ij} = distance between road segment i and road segment j;
m = number of independent variables;
n = number of regulating variables;
$\mathbf{S}$ = vector of dependent variables;
S = average speed of a road, which is a dependent variable;
s_i = average speed of road segment i;
$\mathbf{W}(i)$ = weight matrix corresponding to road segment i;
$\mathbf{W}_j(i)$ = spatial weight of road segment j adjacent to road segment i;
$\mathbf{X}$ = matrix of independent variables with $m'+1$ columns and a column of 1s for the intercept;
x_k = attribute of the built environment, which is an independent variable;
x_{ik} = kth independent variable associated with road segment i;
β_o = constant;
β_{i0} = constant of road segment i;
β_{ik} = regression coefficient corresponding to x_{ik};
β_k = regression coefficient corresponding to x_k;
$\hat{\boldsymbol{\beta}}(i)$ = matrix of regression parameters at road segment i;
ε = random error;
ε_i = random error of road segment i;
η_p = regression coefficient corresponding to D_p; and
λ_{kp} = coefficient of the impact of the interaction between independent variables and regulating variables.

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